

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.</i>		
Student Name:	G. Nithya Sree	Reg. No.:	192424103
Section / Batch:	3 rd Year AIDS	Date:	08/07/2026

Code of Conduct

I G. Nithya Sree(192424103) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Nithya Sree

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
GRAND TOTAL:			100 Marks

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CO1_AT1_C CODE OUTPUT PREDICTION

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QUESTIONS

Q1

C Code – Pointer with Array

10 marks

✓

Q2

C Code - Linked List Node

10 marks

✓

Q3

C Code - Binary Search Complexity

10 marks

Q4

C Code - Pointer Increment

10 marks

Q5

C Code - Array Address

10 marks

Q6

C Code – Linked List Traversal

10 marks

Q1

C Code – Pointer with Array

10 marks

```
#include<stdio.h>

int main(){
    int arr[]={10,20,30};
    int *p=arr;
    printf("%d",*(p+2));
}
```

Your Answer

30

Save Answer

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QUESTIONS

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C Code – Pointer with Array

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✓

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C Code - Linked List Node

10 marks

✓

Q3

C Code - Binary Search Complexity

10 marks

Q4

C Code - Pointer Increment

10 marks

Q5

C Code - Array Address

10 marks

Q6

C Code – Linked List Traversal

10 marks

Q2

C Code - Linked List Node

10 marks

```
#include<stdio.h>

struct Node{
    int data;
    struct Node *next;
};

int main(){
    struct Node n1,n2;
    n1.data=10;
    n2.data=20;
    n1.next=&n2;
    n2.next=NULL;
    printf("%d",n1.next->data);
}
```

Your Answer

20

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QUESTIONS

Q1
C Code - Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q6
C Code - Linked List Traversal
10 marks

Q3 C Code - Binary Search Complexity10 marks

```
while(low <= high){  
    mid=(low+high)/2;  
}
```

Your Answer

O(log n)

Save Answer

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QUESTIONS

Q1
C Code - Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q6
C Code - Linked List Traversal
10 marks

Q4 C Code - Pointer Increment10 marks

```
#include <stdio.h>  
int main(){  
    int a[]={5,10,15};  
    int *p=a;  
    p++;  
    printf("%d",*p);  
}
```

Your Answer

10

Save Answer

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10 marks

✓

Q3

C Code - Binary Search Complexity

10 marks

✓

Q4

C Code - Pointer Increment

10 marks

✓

Q5

C Code - Array Address

10 marks

✓

Q6

✓

Q5 C Code - Array Address

10 marks

```
int arr[]={5,10,15};
printf("%d",*(arr+1));
```

Your Answer

10

Save Answer

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C Code - Linked List Node

10 marks

✓

Q3

C Code - Binary Search Complexity

10 marks

✓

Q4

C Code - Pointer Increment

10 marks

✓

Q5

C Code - Array Address

10 marks

✓

Q6

✓

Q6 C Code - Linked List Traversal

10 marks

```
Node1 -> Node2 -> Node3 (Node values are 10,20,30)
current=head;
current=current->next;
printf("%d",current->data);
```

Your Answer

20

Save Answer

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QUESTIONS

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C Code – Pointer with Array

10 marks

✓

Q2

C Code - Linked List Node

10 marks

✓

Q3

C Code - Binary Search Complexity

10 marks

✓

Q4

C Code - Pointer Increment

10 marks

✓

Q5

C Code - Array Address

10 marks

✓

Q6

✓

Q7 C Code – Recursion (Factorial)

10 marks

```
#include <stdio.h>

int fact(int n){
    if(n==0)
        return 1;
    return n * fact(n-1);
}

int main(){
    printf("%d", fact(4));
    return 0;
}
```

Your Answer

24

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10 marks

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Q2

C Code - Linked List Node

10 marks

✓

Q3

C Code - Binary Search Complexity

10 marks

✓

Q4

C Code - Pointer Increment

10 marks

✓

Q5

C Code - Array Address

10 marks

✓

Q6

✓

Q8 C Code – Recursion (Sum)

10 marks

```
#include <stdio.h>

int sum(int n){
    if(n==0)
        return 0;
    return n + sum(n-1);
}

int main(){
    printf("%d", sum(5));
}
```

Your Answer

15

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QUESTIONS

Q1
C Code - Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q6

Q9 C Code – Primitive Data Type10 marks

```
#include <stdio.h>

int main(){
    int a=10;
    char ch='A';
    printf("%d %c",a,ch);
}
```

Your Answer

10 A

Save Answer

✓ Saved

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QUESTIONS

Q1
C Code - Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q6

Q10 C Code – Linear Data Structure (Array)10 marks

```
#include <stdio.h>

int main(){
    int arr[]={2,4,6,8};
    printf("%d",arr[3]);
}
```

Your Answer

8

Save Answer